

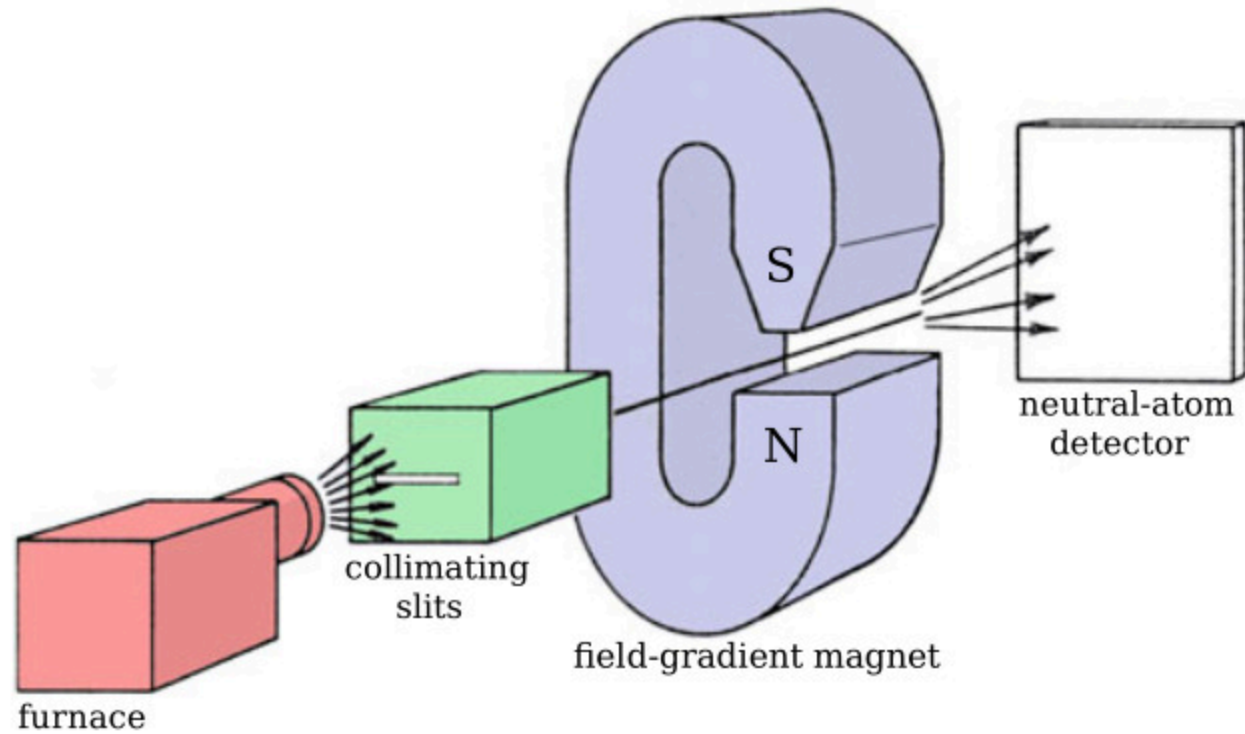
Stern-Gerlach Experiment

Luca Cicu - Lorenzo Liuzzo

27/11/2025

Experiment description

Stern-Gerlach Experiment Apparatus



Angular & Magnetic momenta

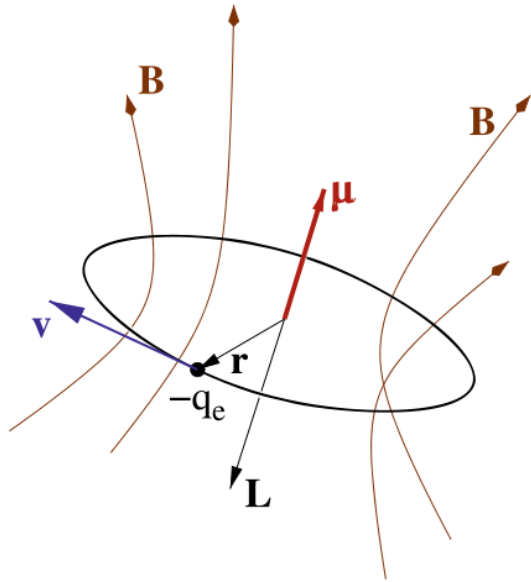
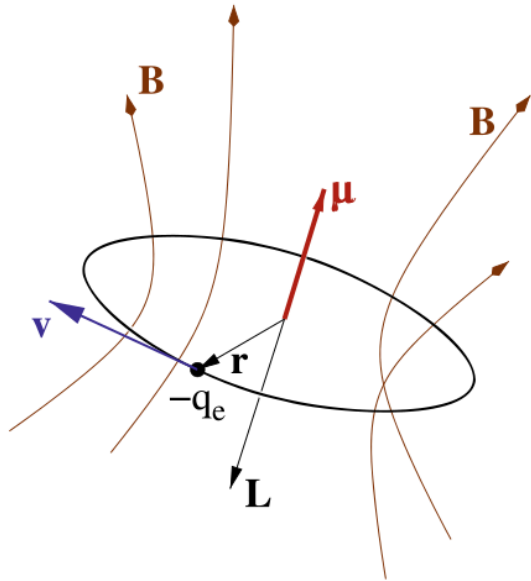


Figure 2: **Angular** and **magnetic** momenta generated by an electron orbiting circularly.

Angular & Magnetic momenta

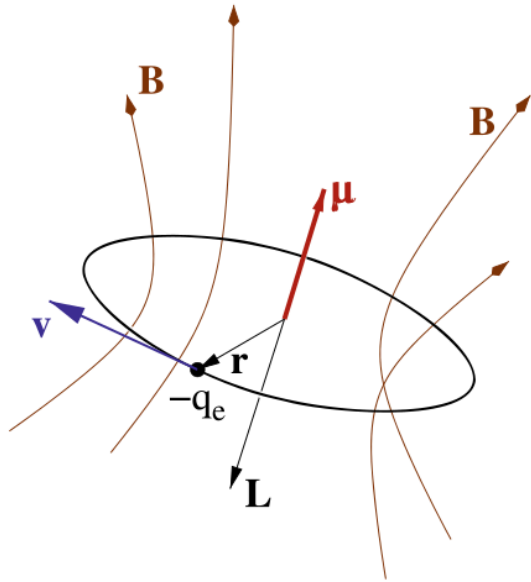


Angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$

Magnetic momentum $\boldsymbol{\mu} = I \Sigma \hat{\mathbf{n}}$

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Angular & Magnetic momenta



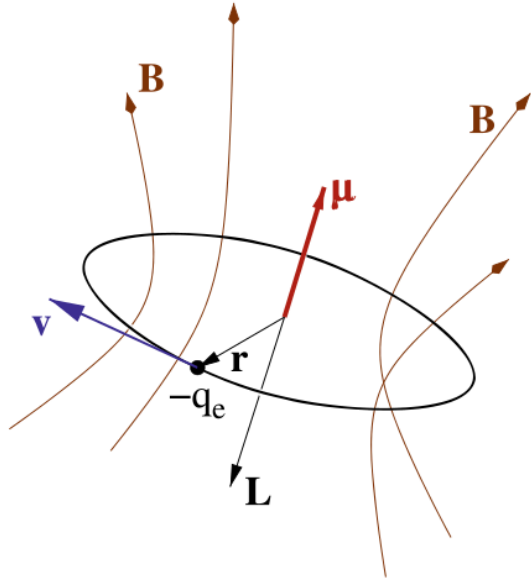
Angular momentum $\mathbf{L} = \mathbf{x} \times \mathbf{p}$

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$$\boldsymbol{\mu} = \frac{qv}{2\pi r} \pi r^2 \hat{\mathbf{n}} = \frac{q}{2m} \mathbf{L}$$

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Stern-Gerlach Experiment

Angular & Magnetic momenta

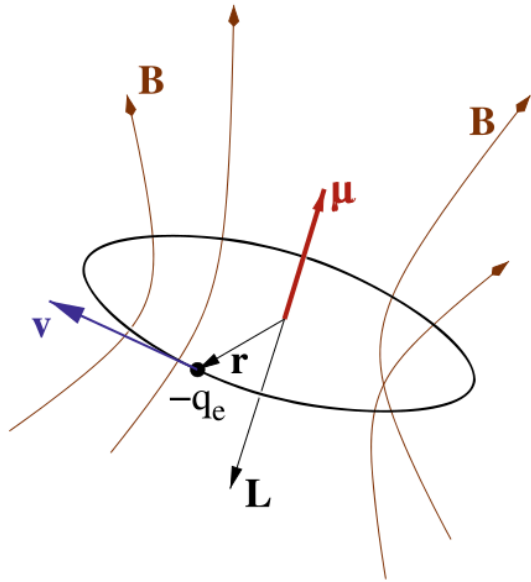


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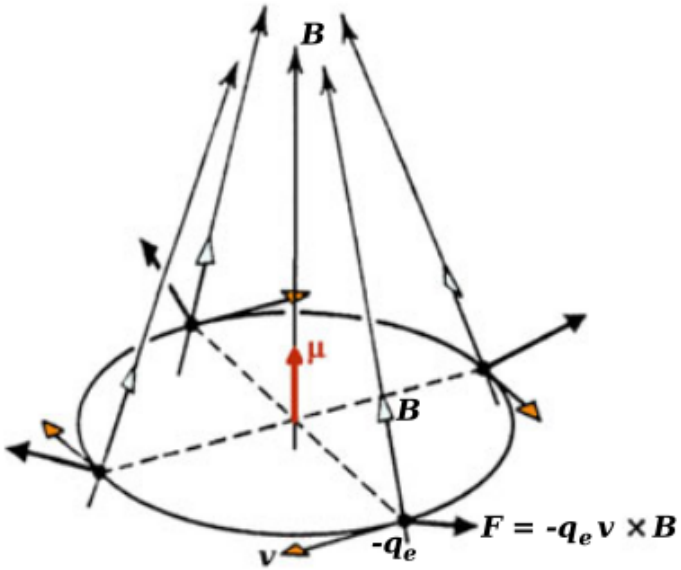
$$\boldsymbol{\mu} = \frac{qv}{2\pi r} \pi r^2 \hat{\mathbf{n}} = \frac{q}{2m} \mathbf{L} \Rightarrow \boldsymbol{\mu} \propto \mathbf{L}$$

The **interaction** between $\boldsymbol{\mu}$ and an **external magnetic field** $\mathbf{B}$ is represented by the magnetic hamitonian

$$H_{\text{magn}} = -\boldsymbol{\mu} \cdot \mathbf{B}$$

Stern-Gerlach Experiment

Magnetic Interaction



$$F_{\text{magn}} = -\nabla H_{\text{magn}} = \nabla(\mu \cdot B)$$

Figure 3: Microscopic origin of the magnetic force.

Stern-Gerlach Experiment

Magnetic Interaction

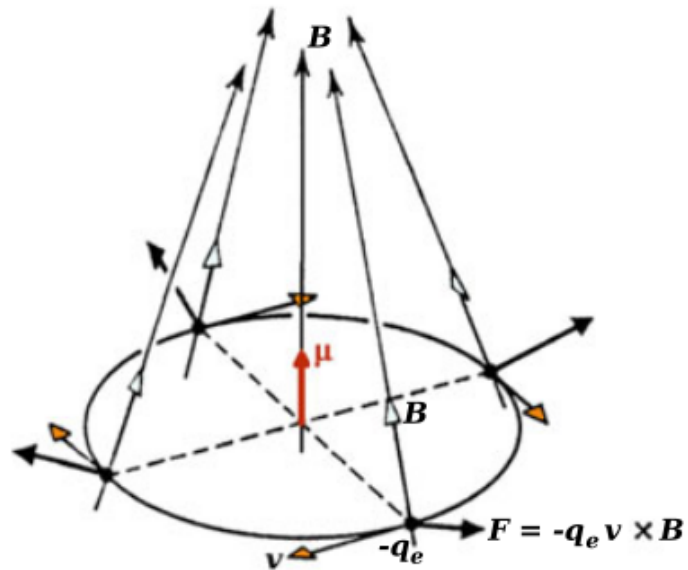


Figure 3: Microscopic origin of the magnetic force.

$$F_{\text{magn}} = -\nabla H_{\text{magn}} = \nabla(\boldsymbol{\mu} \cdot \boldsymbol{B})$$

$$F_z = \mu_z \frac{\partial B_z}{\partial z}$$

Stern-Gerlach Experiment

Magnetic Interaction

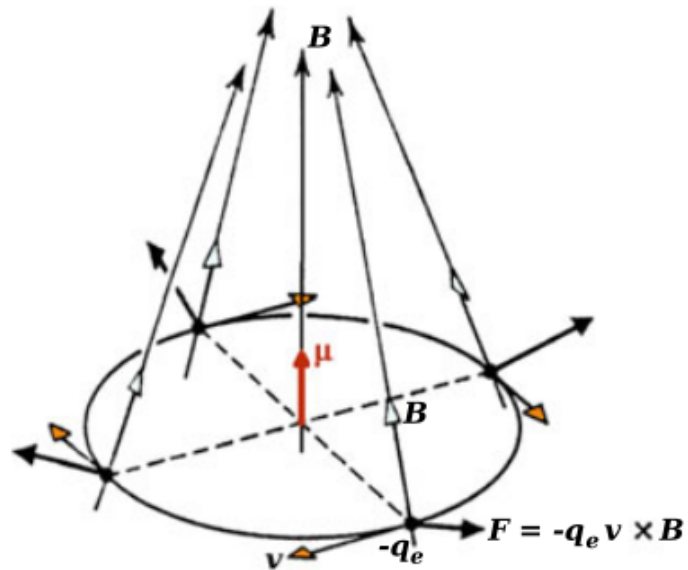


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Stern-Gerlach Experiment

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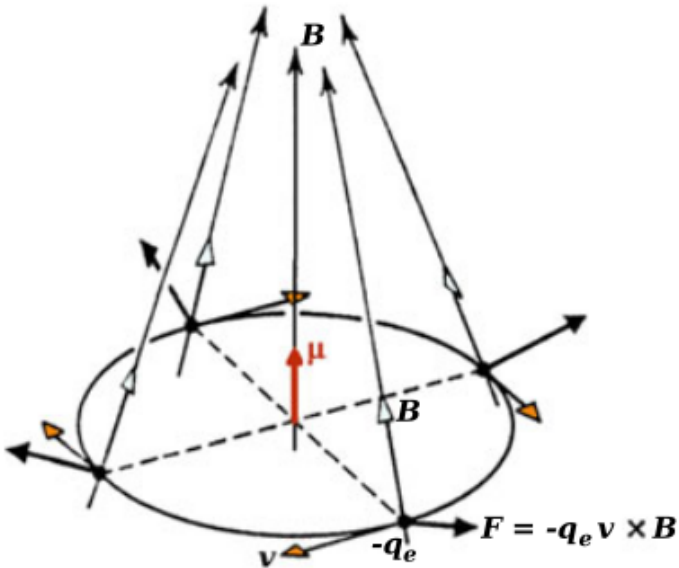


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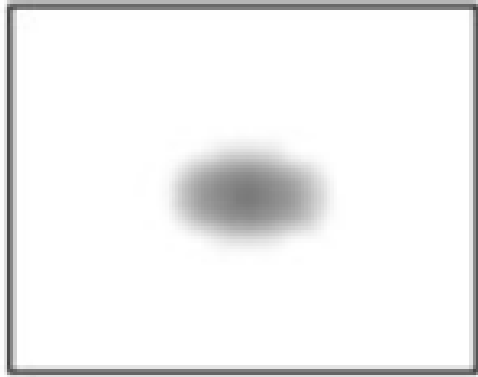
$$F_z = \mu_z \frac{\partial B_z}{\partial z} = m\ddot{z}$$

$$z(t) = z_0 + \dot{z}_0 t + \frac{\mu_z}{2m} \frac{\partial B_z}{\partial z} t^2$$

Experimental Results

Stern-Gerlach Experiment

Classical Results

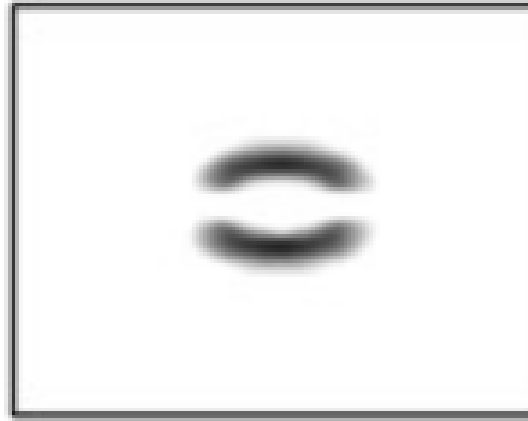


classically predicted

Diocane

Stern-Gerlach Experiment

Observed Results



observed

Why two deflected spots?

Historical Context

Stern-Gerlach Experiment

Historical Context

Planck quanti

Hello world

einstein photoe

Hello world

bohr atom

Hello world

Quantum Mechanics

QM — States

A quantum state is a vector $|\psi\rangle$ in a **Hilbert** space, that is a complete inner product space with normalized elements.

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$$\langle e_i | e_j \rangle = \delta_{ij} \quad I = \sum_i |e_i\rangle \langle e_i|$$

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- **State representation**

$$|\psi\rangle = \sum_i |e_i\rangle \langle e_i | \psi \rangle = \sum_i c_i^\psi |e_i\rangle$$

$c_i^\psi = \langle e_i | \psi \rangle$ are **probability amplitudes**

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- **Inner product**

$$\langle \varphi | \psi \rangle = \sum_i \langle \varphi | e_i \rangle \langle e_i | \psi \rangle = \sum_i (c_i^\varphi)^* c_i^\psi$$

Stern-Gerlach Experiment

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$$\langle \varphi | \psi \rangle = \sum_i \langle \varphi | e_i \rangle \langle e_i | \psi \rangle = \sum_i (c_i^\varphi)^* c_i^\psi$$

- **Born's rule**

$$\langle \psi | \psi \rangle = \sum_i (c_i^\psi)^* c_i^\psi = \sum_i |c_i^\psi|^2 = 1$$

$|c_i^\psi|^2$ is the **probability** to find $|\psi\rangle$ in $|e_i\rangle$

QM — Operators

A **linear operator** $\hat{O}$ is a transformation from an Hilbert space into itself, that acts like

$$\hat{O}(\alpha|\psi\rangle + \beta|\varphi\rangle) = \alpha\hat{O}|\psi\rangle + \beta\hat{O}|\varphi\rangle = \alpha|\psi'\rangle + \beta|\varphi'\rangle$$

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The action of $\hat{O}$ on a state can be entirely represented by its action on some **basis**:

$$|\psi'\rangle = \hat{O}|\psi\rangle = \sum_i \hat{O}|e_i\rangle\langle e_i|\psi\rangle = \sum_{ij} |e_j\rangle\langle e_j|\hat{O}|e_i\rangle\langle e_i|\psi\rangle = \sum_{ij} |e_j\rangle O_{ij} c_i^\psi$$

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We can also define the **adjoint** operator $\hat{O}^\dagger$: $\langle\psi'| = \langle\psi|\hat{O}^\dagger$ $\hat{O}_{ij}^\dagger = (\hat{O}_{ji})^*$

QM — Operators & Observables

We can associate every **observable** of a system to an operator

$$\hat{O} = \sum_i \lambda_i |e_i\rangle \langle e_i| \quad \hat{O}|e_j\rangle = \sum_i \lambda_i |e_i\rangle \langle e_i|e_j\rangle = \lambda_j |e_j\rangle$$

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where λ_i are the **possible results** and $|e_i\rangle$ are the **possible states** in which the system is founded **after** the measurement.

The requirements are that, **independently by the choice of the basis**,

$$\langle e_i|e_j\rangle = \delta_{ij} \quad \lambda_i^* = \lambda_i$$

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Such operators are called **hermitian** and satisfy $\hat{O}_{ij}^\dagger = \lambda_i^* \delta_{ij} = \lambda_i \delta_{ij} = \hat{O}_{ij}$

QM — Commutator of Operators

We define the **commutator** between two operators $\hat{A}$ and $\hat{B}$ as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = -[\hat{B}, \hat{A}]$$

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Two observables can be **simultaneously** determined iff their operators are compatible, that is if they share a **common basis** of eigenstates:

$$\hat{A}|e_i\rangle = \lambda_i|e_i\rangle \quad \hat{B}|e_i\rangle = \mu_i|e_i\rangle \quad \Leftrightarrow \quad \langle e_i | [\hat{A}, \hat{B}] | e_j \rangle = (\lambda_i \mu_i - \lambda_j \mu_j) \langle e_i | e_j \rangle = 0$$

Position and impulse
don't commute!

$$[\hat{p}, \hat{x}] = -i\hbar$$

QM — Angular Momentum

Definition

$$\hat{L}_k = (\hat{\mathbf{x}} \times \hat{\mathbf{p}})_k = \varepsilon_{kij} \hat{x}_i \hat{p}_j$$

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Spectrum

$$\hat{L}^2 |lm\rangle = \hbar^2 l(l+1) |lm\rangle \quad l \in \mathbb{N}$$

$$\hat{L}_z |lm\rangle = \hbar m |lm\rangle \quad -l \leq m \leq l$$

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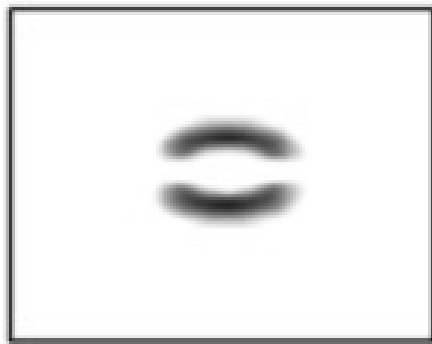
$$\text{Degeneracy} \quad d_l = 2l + 1$$

Theoretical Review

Stern-Gerlach Experiment

Theoretical Review

For an electron, $\hat{\mu} = \frac{\hbar q_e}{2m_e} \frac{\hat{\mathbf{L}}}{\hbar}$

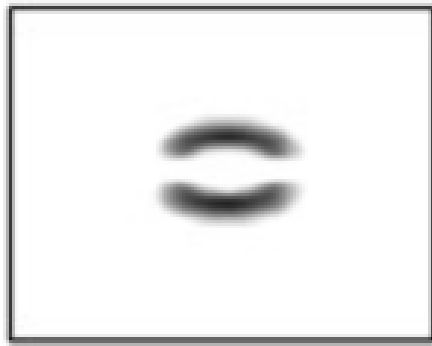


observed

Stern-Gerlach Experiment

Theoretical Review

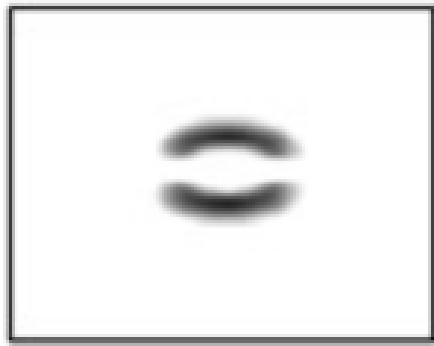
For an electron, $\hat{\mu} = \frac{\hbar q_e}{2m_e} \frac{\hat{\mathbf{L}}}{\hbar} = -g_l \mu_B \frac{\hat{\mathbf{L}}}{\hbar}$



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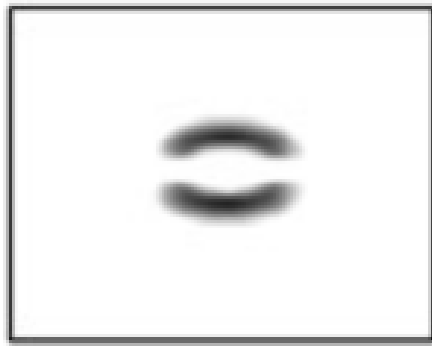


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where μ_B is the **Bohr magneton** and g_l is the **orbital g-factor** (here $g_l = 1$).

Stern-Gerlach Experiment

Theoretical Review



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where μ_B is the **Bohr magneton** and g_l is the **orbital g-factor** (here $g_l = 1$).

The energy of the system $|\psi\rangle = |lm_l\rangle$ is

$$\langle \hat{H} \rangle_\psi = -\langle \hat{\mu}_z B_z \rangle_\psi = g_l \mu_B B_z \frac{1}{\hbar} \langle \hat{L}_z \rangle_\psi$$

$\hat{L}_z$ exhibits an odd number of discrete eigenvalues $(2l + 1)$, so...

Why two deflected spots?

Spin

QM — Intrinsic Angular Momentum

Commutation rules

$$[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k \quad [\hat{S}_i, \hat{S}^2] = 0$$

Basis

The common eigenstates of $\hat{S}^2$ and $\hat{S}_z$ are $|sm_s\rangle$.

Spectrum

$$\hat{S}^2|sm_s\rangle = \hbar^2 s(s+1)|sm_s\rangle$$

$$\hat{S}_z|sm_s\rangle = \hbar m_s|sm_s\rangle \quad -s \leq m_s \leq s$$

Degeneracy $d_s = 2s + 1$

QM — Final Theoretical Review

Consider a state $|\psi\rangle = |lm_l\rangle \otimes |sm_s\rangle$
entering a SG apparatus

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Basis

Since $[\hat{L}_i, \hat{S}_j] = 0$, we can represent $|\psi\rangle$
using the basis $\{\hat{\mathbf{L}}^2, \hat{L}_z, \hat{\mathbf{S}}^2, \hat{S}_z\}$

Energy Spectrum

$$\begin{aligned} E_{m_l m_s} &= \frac{\mu_B B_z}{\hbar} (g_l \langle \hat{L}_z \rangle_\psi + g_s \langle \hat{S}_z \rangle_\psi) \\ &= \frac{\mu_B B_z}{\hbar} (g_l m_l + g_s m_s) \end{aligned}$$

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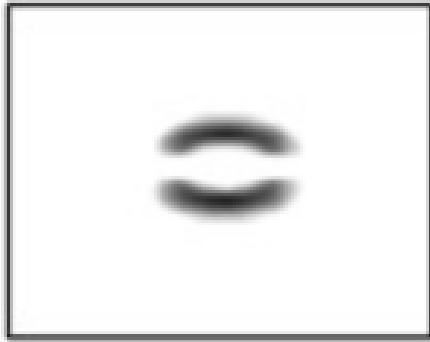
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Degeneracy $d = (2l + 1)(2s + 1)$

Stern-Gerlach Experiment

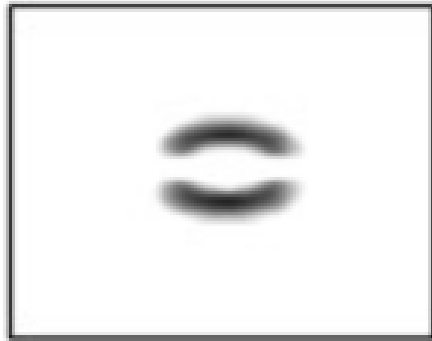
Final Outcomes



observed

Stern-Gerlach Experiment

Final Outcomes



observed

$$d_s = 2 \quad \Rightarrow \quad s = \frac{1}{2} \quad m_s = \pm \frac{1}{2}$$

Stern-Gerlach Experiment

Final Outcomes



observed

$$d_s = 2 \quad \Rightarrow \quad s = \frac{1}{2} \quad m_s = \pm \frac{1}{2}$$

$$\Delta E_{0,\pm\frac{1}{2}} = g_s \frac{\mu_B B_z}{\hbar}$$

Also, to make the deflection magnitude match the experiment $g_s = 2$

Thank you for the attention!

Total Angular Momentum

Consider a state $|\psi\rangle = |lm_l\rangle \otimes |sm_s\rangle$

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Spectrum

$$\hat{\mathbf{J}}^2 |jm_j\rangle = \hbar^2 j(j+1) |jm_j\rangle \quad |l-s| \leq j \leq l+s$$

$$\hat{J}_z |jm_j\rangle = \hbar m_j |jm_j\rangle \quad -j \leq m_j \leq j$$

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$$\hat{\mathbf{J}}^2 = \hat{\mathbf{L}}^2 + \hat{\mathbf{S}}^2 + 2\hat{\mathbf{S}} \cdot \hat{\mathbf{L}}$$

Commutation rules

$$[\hat{L}_i, \hat{S}_j] = 0 \quad [\hat{J}_i, \hat{\mathbf{J}}^2] = 0$$

Basis

The common eigenstates of $\hat{\mathbf{J}}^2$, $\hat{J}_z$ are $|jm_j\rangle$.

Spectrum

$$\hat{\mathbf{J}}^2 |jm_j\rangle = \hbar^2 j(j+1) |jm_j\rangle \quad |l-s| \leq j \leq l+s$$

$$\hat{J}_z |jm_j\rangle = \hbar m_j |jm_j\rangle \quad -j \leq m_j \leq j$$

Degeneracy

$$d_j = 2j + 1$$

Total Angular Momentum

Consider a state $|\psi\rangle = |lm_l\rangle \otimes |sm_s\rangle$

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Degeneracy

$$d_j = 2j + 1 = (2l + 1)(2s + 1)$$

Coupling of Angular Momenta

Consider the magnetic field experienced by an e^- orbiting around a nucleus with Z p^+

$$\mathbf{B}(\mathbf{r}) = \frac{Zq_e}{4\pi\epsilon_0 c^2 m_e} \frac{\hat{\mathbf{L}}}{|\mathbf{r}|^3}$$

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We can represent the e^- state using the basis

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Another option is the **coupled** basis

$$\{\hat{\mathbf{J}}^2, \hat{J}_z, \hat{\mathbf{L}}^2, \hat{\mathbf{S}}^2\}$$

since

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Energy Spectra in Magnetic Field

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{s-o}} + \frac{\mu_B B_z}{\hbar} (\hat{L}_z + 2\hat{S}_z)$$

Energy Spectra in Magnetic Field

Paschen-Back Effect $\mu_B B \gg \xi$

$$\begin{aligned}\xi_{m_l m_s} &= \frac{\mu_B B_z}{\hbar} \left(\langle \hat{L}_z \rangle_\psi + 2 \langle \hat{S}_z \rangle_\psi \right) \\ &= \frac{\mu_B B_z}{\hbar} (m_l + 2m_s)\end{aligned}$$

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Zeeman Effect $\mu_B B \ll \xi$

Features

- Spin-orbit coupling negligible
- L and S precess independently around B
- Good quantum numbers: m_l, m_s
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Landé g-factor

$$g_j = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}$$

Stern-Gerlach Experiment

Boson & Fermions

Hello world

Stern-Gerlach Experiment

Conclusion

Hello world